

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I P.Divakaran(192424260) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue (after visiting node)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Questions:

1. Write the BFS traversal order.

BFS Traversal Order

A → B → C → D → E → F → G

2. Show the queue contents after each iteration.

After visiting A → B, C

After visiting B → C, D, E

After visiting C → D, E, F, G

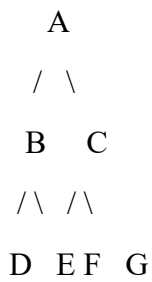
After visiting D → E, F, G

After visiting E → F, G

After visiting F → G

After visiting G → Empty

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack (Top → Left)	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

Perform **Depth-First Search (DFS)** starting from A.

Complete the table.

3. Initial State :

```

1 3 6
5 0 2
4 7 8
  
```

Goal State :

```

1 2 3
4 5 6
7 8 0
  
```

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	[Initial State]	1 0 6 / 5 3 2 / 4 7 8, 1 3 6 / 5 7 2 / 4 0 8, 1 3 6 / 0 5 2 / 4 7 8, 1 3 6 / 5 2 0 / 4 7 8
2	1 0 6 / 5 3 2 / 4 7 8	Remaining 3 states	New child states added to the end of the queue
3	1 3 6 / 5 7 2 / 4 0 8	Remaining queue	New child states added
4	1 3 6 / 0 5 2 / 4 7 8	Remaining queue	New child states added
5	1 3 6 / 5 2 0 / 4 7 8	Remaining queue	New child states added

Questions

1. Which node is expanded in each iteration?

BFS expands the first state in the queue (FIFO order) at each iteration. Starting from the initial state, it removes the front state, generates all valid child states, and adds them to the end of the queue. Since the given initial state is unsolvable, BFS continues expanding states until all reachable states have been explored.

2. How many states are generated in total?

The given initial state cannot reach the specified goal state. Therefore, BFS will eventually generate all reachable states from the initial configuration before stopping. In the 8-puzzle, this is up to 181,440 reachable states.

3. Write the complete solution path from the initial state to the goal state.

No solution path exists, because the given initial state is not solvable for the specified goal state.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores states level by level, where each move has the same cost (one move). Therefore, the first time BFS reaches the goal, it has used the fewest possible moves, guaranteeing the shortest solution in an unweighted 8-puzzle.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

